LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	22
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Mechanical remote controls system: Teleflex, Bowden and use of control tube and cables
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Mechanical remote controls system: Teleflex, Bowden and use of control tube and cables
5. Specific Objectives	17.1.5.1 Trainee will be able to explain the construction and functioning of aircraft flight controls systems and its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Bowden cable tensioning and control tube routing; Safety: Wear gloves to prevent cable pinch injuries; Instructor supervision: Monitor trainee handling of tensioning tools
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a mechanical remote control system?
(b) Name two types of mechanical remote control systems used in aircraft.
(c) Why is a control tube required in a Bowden cable system?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Bowden cable construction and components • Teleflex cable construction and components • Functioning of Bowden cable mechanism • Functioning of Teleflex cable mechanism • Differences between Bowden and Teleflex systems	L & Ds	PPT, IB	15 min
(b)	• Control tube construction and purpose • Cable routing and termination methods • Tensioning procedures and adjustment • Inspection criteria for control cables • Maintenance and lubrication requirements	L & Ds	PPT, PDF	20 min
(c)	• Common failures in remote control systems • Troubleshooting mechanical control issues • Safety considerations during maintenance • Documentation and reporting procedures • Integration with other control systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the control tube in a Bowden cable system.
(b) Describe the key differences between Bowden and Teleflex cable constructions.
(c) List three critical inspection points for aircraft control cables.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of Bowden and Teleflex systems apply to the next lesson on specific control system components (LP 23).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Control system components	Refer to the master syllabus reference list	LP 23

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min